01 / UNEQUAL LINK STUDY

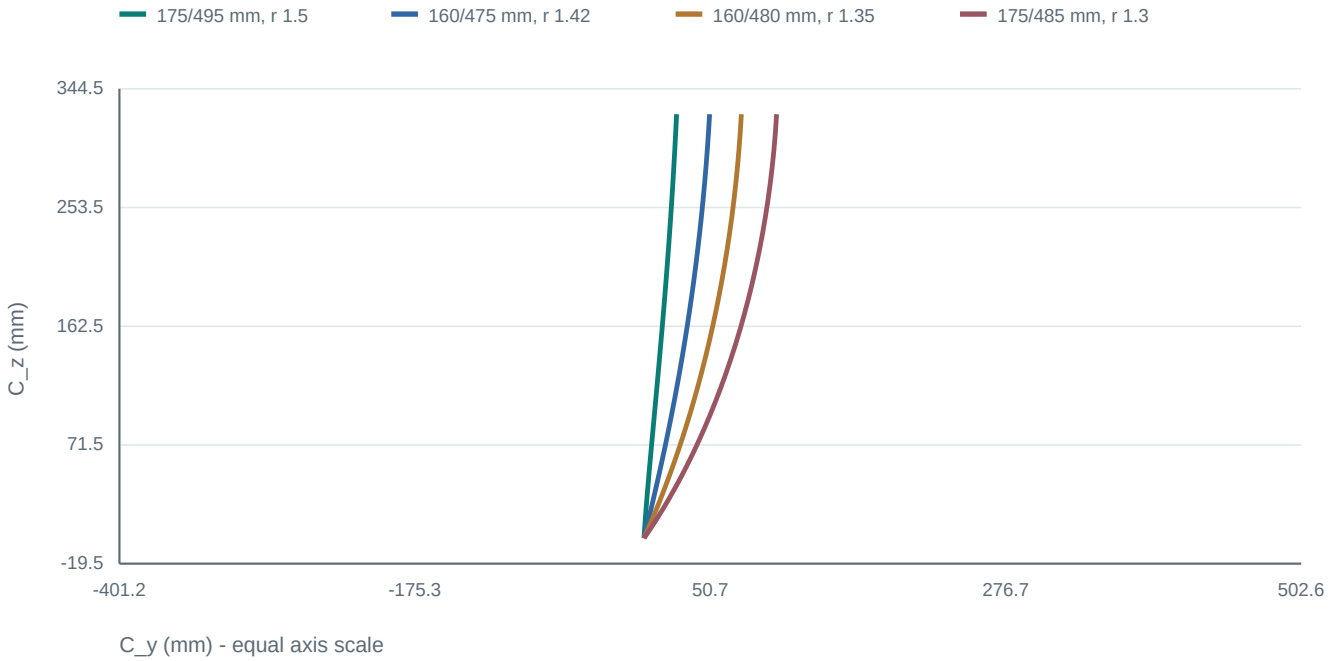
Unequal links that invert

Original alpha=0 and assembly phase=0; wheel radius 200 mm.

A shorter upper link and longer lower link can keep the wheel away from the hip while the height coordinate remains invertible through the fold. These four rounded study examples provide **325 mm total vertical travel, including 20+20 mm assumed reserves: 285 mm useful.**

Upper /lower mm	Guide r	Sideways mm	Hip peak Nm	Hip pitch gap mm
175 /495	1.5	24.90	44.05	30.0
160 /475	1.42	50.16	38.55	29.8
160 /480	1.35	74.45	35.63	39.0
175 /485	1.3	101.41	34.95	32.0

Normal paths aligned at their start; axes use equal physical scales



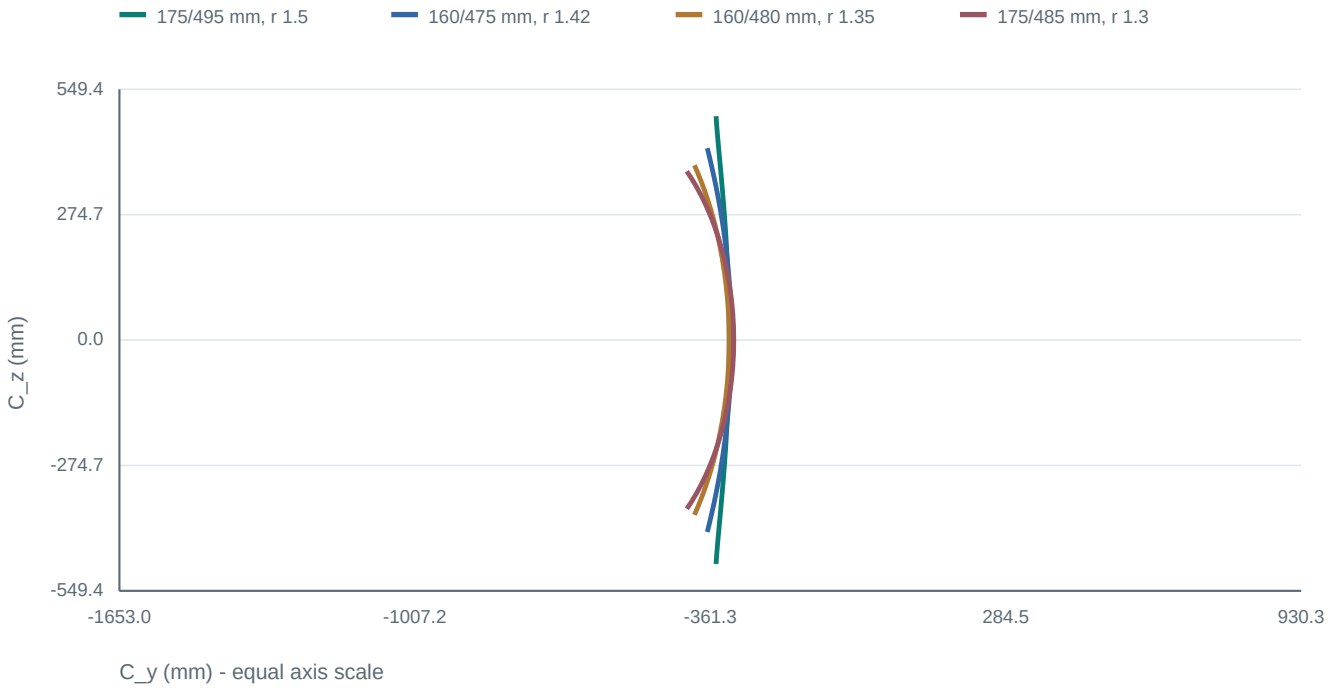
Start with 160/475 mm, r 1.42 for about 50 mm sideways travel. The 160/480 mm, r 1.35 alternative gives about75 mm and lower vertical-load torque. All four pass the assumed wheel/thin-upper-link and wheel/guide-pitch-circle screens. Rod overlap, actual pulley outer size and chassis interference still prevent a claim of physically proven inversion.

02 / UNEQUAL LINK STUDY

Through the fold and back

Full ideal q=100-260 deg; the 325 mm normal stroke occupies only part of this envelope.

Full wheel-centre paths in hip coordinates; equal axis scales



Upper /lower mm	Full height mm	Min Jz mm/rad	Useful hip ROM deg
175 /495	981.0	219.98	49.40
160 /475	840.9	193.95	55.81
160 /480	765.8	176.12	60.16
175 /485	739.2	163.31	61.82

Jz stays positive across the entire 100-260 degree envelope, so each height has one q and an unambiguous branch through 180 degrees. Fold tangent is vertical. Symmetry gives x(-t)=x(t), z(-t)=-z(t): full endpoint sideways drift is zero, although the path bows sideways.

The offline HTML includes four labelled six-second GIFs. Each sweeps 100 to260 degrees and back, shows actual wheel radius and guide pitch circles, and explicitly marks the overlapping fold. They illustrate equations; they do not simulate collision or rolling.

03 / UNEQUAL LINK STUDY

Why the longer lower link helps

Pitch circles improve the screen; they do not establish outer pulley or chassis clearance.

Let a=L2, b=L3, r=ra/rb, t=q-pi and k=r-1. Jonathan conventions give `x=a cos(t)-b cos(kt); z=a sin(t)+b sin(kt)`. Exactly, **$|C|^2=a^2+b^2-2ab\cos(rt)$**, hence minimum wheel-centre/hip distance is |a-b|.

For b>=a the wheel-centre distance to the thin upper segment AB is always at least b-a, with equality at the fold. First screen: b-a>=220 mm (R200 plus 20 assumed gap). With rb 60, the stronger wheel/hip pitch-circle screen is **b-a>=220+60r**. Also require b>=280 for wheel/knee pitch-circle separation and a-60(r+1)>=10 for an assumed 10 mm guide pitch-centre surplus.

160/475 mm, r 1.42: wheel and pitch circles; blue upper /orange lower

Case	Thin upper gap mm	Hip pitch gap mm	Interpretation
160 /475, r 1.42	115.0	29.8	Pass assumed 20 mm screens
150 /375, r1.45	25.0	-62.0	Wheel overlaps hip pitch circle
400 /175, r1.50	-200.0	-65.0	Upper-long wheel overlaps upper rod

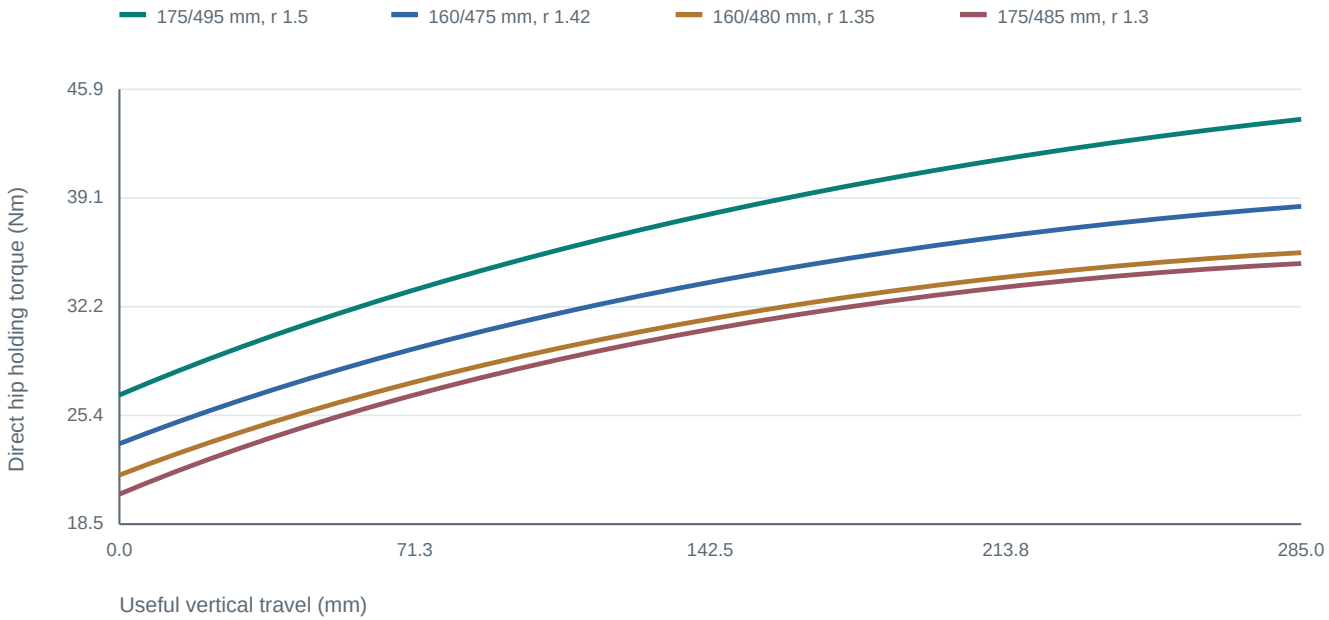
All rods overlap in side view at the fold. For b>a, C lies on the opposite side of A from B and the lower link passes through the hip. Axial staggering, forked links or offset hardware is necessary. The quoted gaps exclude flange/shaft/belt/chassis sizes, rod thickness, tire deformation and tolerances; no candidate is certified collision-free.

04 / UNEQUAL LINK STUDY

Force and control tradeoffs

Pure vertical equivalent corner load 110.3625 N =45 kg x 9.81/4; no spring or moving-link inertia.

Individual direct-hip holding torque curves through 285 mm useful travel



Upper /lower mm	Peak /RMS Nm	Hip speed rad/s	Side-load peak Nm
175 /495	44.05 /37.52	1.035	44.74
160 /475	38.55 /33.17	1.170	39.34
160 /480	35.63 /30.80	1.277	36.42
175 /485	34.95 /30.08	1.353	35.82

Holding torque magnitude is Fz Jz for the plotted upward-load case; reducing its maximum improves worst force per hip Nm, but a smaller Jacobian needs more joint speed. Speed assumes constant vertical 0.25 m/s. Side-load column is the worst either-sign |Fx/Fz|<=0.3 envelope, an assumed sensitivity. For total supported loads of 3g and 4g, multiply the static load term by 3 and 4 respectively. This is not an impact prediction.

Relative-knee torque is **-tau_hip/r**; relative-knee speed is **-r times hip speed**. One-half applies only to r2. Motor gearing is another mapping. Height-RMS equals time-weighted torque RMS for constant vertical speed without dwell. Motor heating still needs shaft/gearing, losses and real duty.

Ideal useful lift work is 31.453 J per leg in every case. Lower torque is therefore mechanical advantage, not improved energy efficiency. Permitted lateral travel also does not establish compliance: local C=J J^T/K_eff needs actual spring/controller and geometric stiffness, contact and rolling conditions. No new motor or spring is selected.

05 / UNEQUAL LINK STUDY

Search priorities and useful ROM

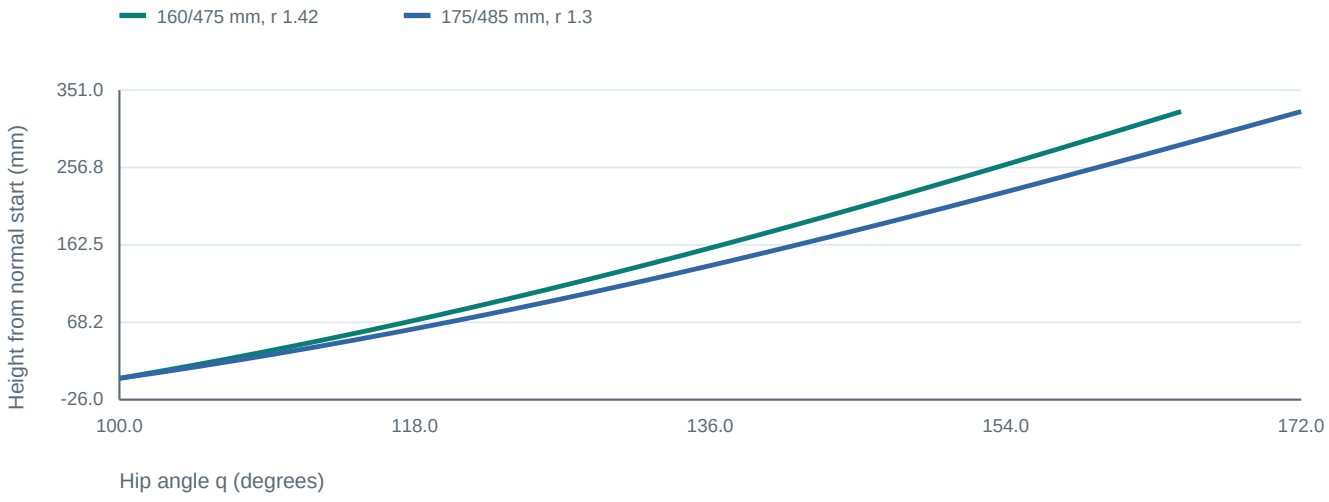
11,535,210 bounded geometry/window combinations; no weighted score or R^2 method.

L2=130-350 and L3=150-650 mm in 5 mm steps; guide 1.2-2.6 in 0.005 steps plus exact 5/3; q start 100-140 in 5 deg steps. End angle solves 325 mm height and must be<=175. Full 100-260 must have Jz>=5 mm/rad. Unequal means |L2-L3|>=10 mm. Normal sideways excursion must be 25/50/75/100 +/-2.5 mm without sideways reversal.

Hard constraints first; geometry-only, thin-segment 220 and hip-pitch-circle families remain separate. Within each, rank minimum useful peak Jz and minimum link sum separately. Coarse 33 poses retain eight candidates per bucket; finalists get continuous derivative-extrema checks and 2001-pose force integrals. Finite grid/pool retention cannot prove a global or manufacturing optimum.

Comparison	L2 /L3, r	Hip peak Nm	Hip pitch gap mm
50 mm grid force leader	155 /460, 1.385	36.43	21.9
50 mm rounded example	160 /475, 1.42	38.55	29.8
100 mm grid size leader	170 /470, 1.29	33.56	22.6
100 mm grid force leader	160 /615, 1.23	33.05	181.2

Height reserves trim different angular windows



Useful ROM uses exact height roots qlow=z^-1(zstart+20), qhigh=z^-1(zend-20), rather than subtracting fixed degrees. For 160/475,r 1.42, totalq=100-164.692 deg; useful ROM 55.806 deg. Rounded examples trade a little peak torque for larger assumed pitch-circle gaps. Starting angles and all margins are study assumptions.

06 / UNEQUAL LINK STUDY

Checks and next design decision

Numerical agreement supports the equations; physical inversion still needs a packaged design.

22 numerical tests pass, including six new unequal-link tests. 8,184 scalar comparisons across 22 cases agree with original Jonathan V2 and Andy 2D statics, maximum error 2.842e-13 mm/Nm. The first test run misclassified diagnostic wide_reverse as a grid ID; the harness classifier was fixed and all 22 tests rerun. Numerical acceptance criteria were unchanged.

Actual Pymunk case	2 ms mm	1 ms mm	0.5 ms mm	0.25 ms mm
clear25	0.05524	0.02742	0.01366	0.00682
clear50	0.04793	0.02380	0.01186	0.00592
clear100	0.06540	0.03248	0.01619	0.00808

All three fixtures pass path/pin/guide/tracking tolerances at all four timesteps and converge. Real motor, pivot and gear constraints move positive-mass bodies; only the initial pose is assigned. Zero gravity/forces and no collision shapes make this a geometry test. It does not validate loaded torque, ground contact, impacts or packaging.

Recommended next CAD seed: 160/475 mm, guide1.42. It provides 50.165 mm normal sideways travel, 38.552 Nm equivalent 45 kg hip peak and 29.8 mm minimum wheel-to-hip pitch-circle gap. Use 160/480,r 1.35 if 74.451 mm sideways travel is desirable: 35.633 Nm peak and 39 mm pitch gap. Both require an explicit axial layout at the fold.

Next measure pulley outer radii, rod/fork thickness, axial stack, chassis envelope and actual contact conditions. Then size springs/motors using that geometry and measured mass/inertia/duty. Prior spring cases remain separate. Historical dual-rope stays **IMPOSSIBLE /rejected**; no Motion Lab controls or services change.

Rebuild using unequal_inversion_search.py, test_unequal_inversion, scripts/verify_unequal_geometry.py, scripts/verify_unequal_report.cjs --math-only, scripts/build_unequal_visuals.py and scripts/build_unequal_report.py. Then run the full browser verifier. Adjacent design record retains commands, source hashes and decisions. Source base 1a3fb7e; workspace scope; canonical physics NOT APPLICABLE.